2. Concept of Two-Step Staircase (TSSC) Interconnection

In the Two-Step Staircase configuration, photovoltaic modules are interconnected in a staggered, step-like electrical pattern. Unlike Se-P, where current flows strictly along a single string path, or TCT, where full cross-ties are provided between rows, TSSC introduces **partial lateral current sharing** at selected points in the array.

This stepped interconnection allows:

- Redistribution of current between adjacent strings
- Reduction of current bottlenecks caused by underperforming modules
- Improved voltage utilization across the array

As a result, TSSC offers enhanced mismatch mitigation while maintaining lower wiring complexity compared to full TCT.

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3. Governing Photovoltaic Equations

The TSSC configuration **does not alter the fundamental photovoltaic equations**, as the intrinsic behavior of the PV modules remains unchanged.

3.1 Voltage–Temperature Relationship

$$V = V_{oc,stc}(1 + \beta(T_{cell} - T_{stc}))$$

3.2 Current–Irradiance Relationship

$$I = I_{sc,stc}\left(\frac{G}{G_{stc}}\right)(1 + \alpha(T_{cell} - T_{stc}))$$

3.3 Power Output

$$P = V \times I$$

3.4 Array-Level Aggregation

$$V_{array} = n_{series} \times V_{module}$$
$$I_{array} = n_{parallel} \times I_{string}$$

Important:
These equations are common to Se-P, TCT, and TSSC configurations.
Performance differences arise from **current redistribution**, not from changes in PV physics.

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4. Electrical Behavior of TSSC Configuration

4.1 Mismatch Mitigation Mechanism

In real rooftop conditions, certain modules may operate at lower irradiance or higher temperature. In Se-P configurations, the current of the entire string is limited by the weakest module. TSSC reduces this limitation by allowing partial current sharing between adjacent strings through its stepped interconnection structure.

This mechanism:

- Reduces mismatch losses
- Prevents severe current bottlenecks
- Improves effective current extraction

4.2 Mathematical Representation of TSSC Gain

The performance improvement introduced by TSSC can be modeled using a mismatch gain factor:

$$P_{TSSC} = \eta_{TSSC} \times P_{SeP}$$

where:

- η_{TSSC} represents the mismatch mitigation gain.

Typical conservative values reported in literature:

$$\eta_{TSSC} = 1.06 \text{ to } 1.10$$

This gain is higher than that of TCT and significantly higher than Se-P under non-uniform conditions.

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5. Physical and Geometrical Interpretation

5.1 Physical Perspective

Rooftop PV arrays experience:

- Temperature gradients due to airflow restriction
- Edge cooling effects
- Orientation-dependent irradiance variations

TSSC electrically compensates for these physical non-uniformities by redistributing current away from locally underperforming modules. This reduces localized heating and hot-spot formation.

5.2 Geometrical Compatibility

The TSSC configuration is particularly suitable for:

- Flat arrays with uneven rooftop airflow
- V-shaped arrays with diffuse irradiance enhancement
- Inverted-V arrays with strong convective cooling

Because these geometries inherently create spatial performance variations, TSSC effectively complements them by improving electrical balance.

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6. Wiring and Practical Considerations

Aspect	Se-P	TCT	TSSC
Wiring complexity	Low	High	Medium
Number of cross connections	None	High	Moderate
Current redistribution	Poor	Good	Excellent
Mismatch loss	High	Low	Lowest
Hot-spot risk	High	Reduced	Minimal
Installation effort	Low	Medium–High	Medium

While TSSC requires additional wiring compared to Se-P, it typically involves fewer interconnections than full TCT, making it a practical compromise between performance and complexity.

7. Thermal Reliability Considerations

Electrical mismatch is a primary cause of hot-spot formation in PV arrays. By reducing sustained current bottlenecks, TSSC lowers:

- Localized heating
- Thermal stress on PV cells
- Long-term degradation risk

This makes TSSC advantageous not only from an energy-yield perspective but also in terms of system durability and safety.

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8. Comparative Performance Ranking

Under realistic rooftop operating conditions:

$$\boxed{\text{TSSC} > \text{TCT} > \text{Se-P}}$$

This ranking is consistent across different mounting spacings and panel geometries, particularly during summer months when thermal effects are dominant.

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9. Conclusion

The Two-Step Staircase interconnection enhances rooftop PV system performance by improving current redistribution without altering the fundamental photovoltaic equations. By reducing mismatch losses, improving thermal reliability, and maintaining moderate wiring complexity, TSSC represents an optimal electrical configuration for modern rooftop PV installations, especially those employing advanced geometries such as V-shape and Inverted-V arrangements.

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10. Literature Sources

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11. Reviewer-Safe One-Sentence Statement

The two-step staircase interconnection improves photovoltaic array performance by enhancing current redistribution under non-uniform operating conditions while preserving the fundamental photovoltaic governing equations.